This document is written with the intention of helping the author learn. Not only by teaching, but also in the scenario where a time traveler gifts it to their younger self. Other readers are also welcome, but be warned: this text is written for highly motivated, mathematically mature readers. Readers are expected to be able to supply rigorous proofs where they are omitted, try their hand at proving results before reading the given proofs, internalize the given proofs in their own terms, work out cases on their own, and so on. The author believes the best way to learn is with a mentor. The author knows there are easily available books far better than this document, in every respect, except that it lets the author do as they please.

Number Theory - Lecture 1 - The Fundamental Theorem of Arithmetic

Fact (Principle of Induction)**.** *Any non-empty subset of the positif integers contains a smallest element.*

Fact + Nomenclature. *Let d be a positive integer and m be an integer. Then, there exists a unique pair of integers q, r satisfying $m = dq + r$ and $0 \leq r < d$. We call q the quotient and r the remainder of the division of m by d. When $r = 0$, that is, when there exists an integer q with $m = dq$, we say that d divides m, that d is a divisor of m, or that m is divisible by d, and we write this relation as $d \mid m$.*

Nomenclature. *An integer $p \geq 2$ is called a prime number if its only divisors are 1 and p.*

Fact. *Each positive integer $n \geq 2$ is divisible by at least one prime number p.*

Fact (Primes don't materialize out of thin air)**.** *Let p be a prime number, and let m, n be integers satisfying $p \mid mn$. Then, either $p \mid m$ or $p \mid n$.*

Proof. Without loss of generality, we may restict our attention to positif m, n. By the principle of induction, we may suppose the theorem holds for all prime numbers p' smaller than p (and any n, m). Again by the principle of induction, we may suppose the theorem holds for all positif pairs m', n' with a smaller sum $m' + n' < m + n$ (and our fixed p). Applying division with remainder, we may suppose $0 \leq m, n < p$. We may now write $kp = mn$ for some integer $0 \leq k < p$.
Case 1. If $k = 0$ then either $m = 0$ or $n = 0$, and we have our desired conclusion.
Case 2. We cannot have $k = 1$, by p being a prime number.
Case 3. If $k \geq 2$, then we may pick some prime number q dividing k. In particular $q \mid mn$ and $q < p$. Thus, the (first) induction hypothesis, yields that either $q \mid m$ or $q \mid n$. Without loss of generality, we may assume $q \mid m$. We may write $k = qd$ and $m = qr$ for some positif integers d, r with $r < m$, yielding $dp = rn$. Thus, the (second) induction hypothesis yields that either $p \mid r$ (implying $p \mid m$) or $p \mid n$, reaching our desired conclusion. $\square$

Fact (Fundamental Theorem of Arithmetic)**.** *Each positive integer n may be written as a product of a list of prime numbers $n = p_1 \ldots p_\ell$, in a manner unique up to reordering of the terms.*

Number Theory - Exercises for Lecture 1

Fact. *$d \mid m$ and $m \mid n$ implies $d \mid n$.*

Fact. *m and n have the same remainder with respect to division by d if and only if $d \mid m - n$.*

Fact. *The remainders of $m + n$ and of mn with respect to division by d depends only on the remainders of m and of n with respect to division by d.*

Fact + Nomenclature. *For a positive integer n and a prime number p, let the p-adic valuation $\nu_p(n)$ be the number of times we may divide n by p (until the result is no longer divisible by p). We have*

$$n = \prod_p p^{\nu_p(n)}$$

Fact. *We have $\nu_p(nm) = \nu_p(n) + \nu_p(m)$.*

Fact. *We have $\nu_p(n + m) \geq \min(\nu_p(n), \nu_p(m))$, and equality holds if $\nu_p(n) \neq \nu_p(m)$ (but not necessarily otherwise).*

Fact. *A positive integer is a perfect k-th power if and only if $k \mid \nu_p(n)$ for all prime numbers p. Deduce that the square root of 2 is irrational. More generally, the k-th root of a positive integer which is not a perfect k-th power, is irrational.*

Fact. *The number of divisors of n equals $\prod_p (\nu_p(n) + 1)$. In particular, n has an odd number of divisors if and only if n is a perfect square - convince a child of this.*

Exercise. *Search for Zarmelo's proof of the Fundamental Theorem of Arithmetic, which does not make use of the fact $p \mid mn \implies p \mid m$ or $p \mid n$.*

TODO: fix lecture 1 given the following.

Nomenclature. *We call a* <u>*invertible*</u> *modulo n if $0, a, 2a, 3a, \ldots, (n-1)a$ cover all n distinct residue classes modulo n. Namely, if the row corresponding to a in the multiplication table modulo n contains each residue precisely once.*

Fact. *a is invertible modulo n if and only if there exists b such that $ab \equiv 1 \mod n$. In that case, b is unique modulo n.*

Fact. *xy is invertible modulo n if and only if both x and y are invertible modulo n.*

Fact. *Let p be a prime number. Then each non-zero residue a is invertible modulo p.*

Proof. By induction on $a = 1, \ldots, p-1$. For $a = 1$ the result is clear. Assume the result holds for all previous a values. Divide p by a with remainder so that $p = aq + r$ for some $0 \leq r < a$. Since p is a prime number and $1 < a < p$, we have that $a \nmid p$ so $1 \leq r < a$, and thus the inductive hypothesis yields that $r \equiv -aq \mod p$ is invertible modulo p, and hence so is a. $\square$

Nomenclature. *The number of invertible residue classes modulo n is denoted $\varphi(n)$.*

Fact. *The multiplication table modulo n, reduced to the rows and columns of invertible residues modulo n (so that it has size $\varphi(n) \times \varphi(n)$), has only entries that are invertible modulo n, and moreover each row (and column) contains each invertible residues precisely once.*

Fact. *For each invertible residue class a modulo n we have $a^{\varphi(n)} \equiv 1 \mod n$.*

Proof. Let $x_1, \ldots, x_{\varphi(n)}$ list the invertible residue classes modulo n. Then $ax_1, \ldots, ax_{\varphi(n)}$ is a reordering of the same list. Therefore, the products of the two lists are equal, that is $x_1 \ldots x_{\varphi(n)} \equiv ax_1 \ldots ax_{\varphi(n)} = a^{\varphi(n)} x_1 \ldots x_{\varphi(n)} \mod n$. Since $x_1, \ldots, x_{\varphi(n)}$ are invertible modulo n, we have $1 \equiv a^{\varphi(n)} \mod n$ as desired. $\square$

Quadratic Residues and Quadratic Forms

Fact (Thue)**.** *Let a be coprime to n, and $n \geq 2$. Then, there exists a solution to $ax \equiv y \mod n$ with $0 < |x|, |y| < \sqrt{n}$.*

Proof. Consider all combinations $ax - y \mod n$ where $x, y = 0, \ldots, \lfloor \sqrt{n} \rfloor$. There are $(1 + \lfloor \sqrt{n} \rfloor)^2 > n$ such combinations, so by the pigeon hole principle there must be a repetition. So let $ax_1 - y_1 \equiv ax_2 - y_2$ where $(x_1, y_1) \neq (x_2, y_2)$. Since a is coprime to n, we must have that $x_1 \not\equiv x_2$ and $y_1 \not\equiv y_2$. Now $x = x_1 - x_2$ and $y = y_1 - y_2$ satisfy the requested requirments, as desired. $\square$

Fact. *Let $p \equiv 1 \mod 4$. Then, there exists x, y with $p = x^2 + y^2$.*

Proof. Let a solve $a^2 \equiv -1 \mod p$. Applying Thue's result, let x, y solve $ax \equiv y \mod p$ with $0 < |x|, |y| < \sqrt{p}$. Then $0 < x^2 + y^2 < 2p$ and $x^2 + y^2 \equiv x^2(1 + a^2) \equiv 0 \mod p$, so $p = x^2 + y^2$ as desired. $\square$

Algebraic Integers

Fact. *A number which is both rational and an algebraic integer must be an integer.*

$$\mathbf{Q} \cap \overline{\mathbf{Z}} = \mathbf{Z}$$

Fact (Fundamental Theorem of Symmetric Polynomials)**.** *Let $f(x_1, \dots, x_n)$ be a symmetric polynomial. Then, in fact, $f(x_1, \dots, x_n) = g(e_1, \dots, e_n)$ may be given as a polynomial expression in the elementary symmetric polynomials in $x_1, \dots, x_n$.*

Proof. We give a lexicographic ordering of monomials, so that $x_1^{i_1} \dots x_n^{i_n} >_{lex} x_1^{j_1} \dots x_n^{j_n}$ whenever $i_k > j_k$ for the first index k where $i_k \neq j_k$. This is a total order of all monomials. Suppose the lexicographically highest monomial term of $f(x_1, \dots, x_n)$ is $x_I = x_1^{i_1} \dots x_n^{i_n}$, and that the theorem holds for all symmetric polynomials containing terms only lexicographically smaller than this monomial. Since f is symmetric, we have $i_1 \geq \dots \geq i_n$. Now, $E_I = e_1^{i_1 - i_2} e_2^{i_2 - i_3} \dots e_{n-1}^{i_{n-1} - i_n} e_n^{i_n}$ has highest lexicographic monomial term equal to $x_1^{i_1 - i_2} (x_1 x_2)^{i_2 - i_3} \dots (x_1 \dots x_{n-1})^{i_{n-1} - i_n} (x_1 \dots x_n)^{i_n} = x_I$. Subtracting cE_I from f (where c is the coefficient of x_I in f) yields a symmetric polynomial, all of whose terms are lexicographically smaller than x_I, so by induction $f - cE_I$ is expressible as a polynomial in $e_1, \dots, e_n$, and thus so is f, as desired. $\square$

Fact. *Sums and products of algebraic integers are algebraic integers.*

Proof. Let α, β be algebraic integers. Then we may write

$$(x - \alpha_1) \dots (x - \alpha_m) = x^m + a_1 x^{m-1} + \dots + a_{m-1} x + a_m$$

$$(x - \beta_1) \dots (x - \beta_n) = x^n + b_1 x^{n-1} + \dots + b_{n-1} x + b_n$$

for integers a_i, b_j and numbers α_i, β_j with $\alpha = \alpha_1$, $\beta = \beta_1$. Consider the polynomials

$$\prod_{i,j} (x - (\alpha_i + \beta_j))$$

and

$$\prod_{i,j} (x - (\alpha_i \beta_j))$$

By the fundamental theorem of symmetric polynomials, each of these is expressible as a polynomial in x whose coefficients are polynomials in a_i, b_j. That is, these polynomials have integer coefficients. In particular, $\alpha + \beta$ and $\alpha\beta$ are algebraic integers, as desired. $\square$

Proof. Let $A \in \mathbf{Z}^{m \times m}$, $B \in \mathbf{Z}^{n \times n}$ be respectively the companion matrices of α, β. Let A have eigenvalues α_i with corresponding eigenvectors v_i, and let B have eigenvalues β_j with corresponding eigenvectors u_j. Then $A \otimes B \in \mathbf{Z}^{nm \times nm}$ and $A \otimes \mathrm{id}_n + \mathrm{id}_m \otimes B$ have respectively the eigenvalues $\alpha_i \beta_j$ and $\alpha_i + \beta_j$, both with corresponding eigenvectors $v_i \otimes u_j$. It follows that $\alpha_i \beta_j$ and $\alpha_i + \beta_j$ (and in particular $\alpha\beta$ and $\alpha + \beta$) are algebraic integers, as desired. $\square$

Cyclotomics

Nomenclature. *The n-th cyclotomic polynomial is*

$$\Phi_n(x) = \prod_{\mu \text{ a primitive } n\text{-th root of unity}} (x - \mu) = \prod_{j \in (\mathbf{Z}/n\mathbf{Z})^\times} (x - \zeta^j)$$

where $\zeta = e^{2\pi i/n}$.

Fact.

$$x^n - 1 = \prod_{d|n} \Phi_d(x)$$

Fact. $\Phi_n(x) \in \mathbf{Z}[x]$ *is a monic integer polynomial.*

Proof. Clearly $\Phi_n(x)$ is a monic polynomial. By induction, we have that $\Phi_n(x)$ is a rational function of x with rational coefficients, but it has no poles so it is a rational polynomial. $\Phi_n(x) \in \mathbf{Q}(x) \cap \mathbf{C}[x] = \mathbf{Q}[x]$. It is clear that the roots of $\Phi_n(x)$ are algebraic integers, so the coefficients of $\Phi_n(x)$ are algebraic integers that are rational numbers, and so $\Phi_n(x)$ has integer coefficients. $\Phi_n(x) \in \mathbf{Q}[x] \cap \overline{\mathbf{Z}}[x] = \mathbf{Z}[x]$. $\square$

Fact. $\Phi_n(x)$ *is irreducible in* $\mathbf{Q}[x]$.

Proof. (Schur) Recall that the discriminant of $x^n - 1$ is $\pm n^n$. Let $f(x)$ be the minimal polynomial of ζ over $\mathbf{Q}$. So $f(x) \in \mathbf{Z}[x]$ and $f(x) = (x - \zeta_1) \dots (x - \zeta_k)$ for some distinct n-th roots of unity $\zeta_1, \dots, \zeta_k$. We must show that the roots of $f(x)$ are closed under exponentiating by powers coprime to n. It suffices to show that if μ is a root of f and p is a prime coprime to n then μ^p is a root of f. We have that $f(\mu^p) = f(\mu^p) - f(\mu)^p = f(x^p) - f(x)^p \restriction_{x=\mu} = pg(x) \restriction_{x=\mu}$ for some $g(x) \in \mathbf{Z}[x]$, so that $p \mid f(\mu^p)$ over $\overline{\mathbf{Z}}$. Moreover, if we assume for the sake of contradiction that μ^p is not a root of f then $f(\mu^p) = (\mu^p - \zeta_1) \dots (\mu^p - \zeta_k)$ divides the discriminant $\pm n^n$ over $\overline{\mathbf{Z}}$. However, $p \mid \pm n^n$ over $\overline{\mathbf{Z}}$ implies $p \mid n$ over $\mathbf{Z}$, contrary to our assumption. This shows that μ^p is a root of f, as desired. $\square$